YOUNG SU KO

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EDUCATION

University of California, San Diego La Jolla, CA
Ph.D. Biochemistry and Molecular Biophysics (GPA: 4.0/4.0) Expected September 2027
Dissertation: AI-guided Design and Prediction of Protein Interactions

Pomona College Claremont, CA
B.A. Chemistry, *magna cum laude* (GPA: 3.99/4.0) May 2022

RESEARCH EXPERIENCE

University of California, San Diego January 2023 – Present
Graduate Student Researcher, Advisor: Wei Wang

- Designed forge, a sequence-based generative model for protein binder design using flow matching over protein language model representations, enabling structure-free protein design.
- Applied AI-driven sequence optimization to engineer an EGFR-binding protein with 67% stronger affinity than native EGF in the Adaptyv Bio binder design competition.
- Developed TUnA, a Transformer-based model for protein-protein interaction prediction, achieving state-of-the-art AUROC (0.70) on a leakage-free benchmark.
- Analyzed representation transfer by benchmarking six protein language models across six datasets; proposed an embedding fusion framework to leverage multiple representations.

Pomona College September 2019 – May 2022
Undergraduate Researcher, Advisor: Roberto Garza-Lopez

- Performed virtual screening of SARS-CoV-2 main protease inhibitors using molecular docking and molecular dynamics simulations.

PUBLICATIONS, PREPRINTS, AND WORKSHOPS

Ryan Hard, Jonathan Parkinson, **Young Su Ko**, Wei Wang. “Antibody Affinity Maturation by Yeast Mating Against a Diverse Antigen Library Enhances Antibody Resistance to Viral Mutations.” ***ACS Pharmacology & Translational Science (in press)*** (2026).

M. Frank Erasmus, …, Jonathan Parkinson, **Young Su Ko**, Wei Wang, …, Andrew R.M. Bradbury. “A blinded, prospective benchmark of in silico antibody discovery anchored to experimental affinity and developability.” ***Nature Biotechnology*** (2026).

Kapil Devkota, Daichi Shonai, Joey Mao, **Young Su Ko**, Wei Wang, Scott Soderling, Rohit Singh. “Miniaturizing and modifying natural proteins with Raygun.” ***Nature*** (2026).

Li Zhang, Han Guo, Leah Schaffer, **Young Su Ko**, Digvijay Singh, Hamid Rahmani, Danielle Grotjahn, Elizabeth Villa, Michael Gilson, Wei Wang, Trey Ideker, Eric Xing, Pengtao Xie. “ProteinAligner: A Multi-modal Pretraining Framework for Protein Foundation Models.” ***Cell Reports Methods*** (2026).

Young Su Ko*, Jonathan Parkinson, Wei Wang. “Scalable embedding fusion with protein language models: insights from benchmarking text-integrated representations.” ***Briefings in Bioinformatics*** (2026).

Young Su Ko*, Wei Wang. “forge: sequence-based binder design with latent flow matching.” ***Machine Learning in Structural Biology (MLSB)*** (2025).

Jonathan Parkinson, Ryan Hard, **Young Su Ko**, Wei Wang. “RESP2: An uncertainty-aware multi-target multi-property optimization AI pipeline for antibody discovery.” ***Advanced Science*** (2025).

Young Su Ko*, Jonathan Parkinson, Cong Liu, Wei Wang. “TUnA: an uncertainty-aware transformer model for sequence-based protein–protein interaction prediction.” ***Briefings in Bioinformatics*** (2024).

SKILLS

Engineering: Python, PyTorch, PyTorch Lightning, Numpy, Hydra, Weights & Biases, Hugging Face, Slurm, Git, Linux
Research: Generative modeling (flow matching, diffusion), protein language modeling, protein design workflows (RFDiffusion, ProteinMPNN, AlphaFold / Boltz)

PRESENTATIONS

Poster: *forge: sequence-based binder design with latent flow matching*. **Machine Learning in Structural Biology (MLSB)**, San Diego, USA, December 2025.

Poster and Lightning Talk: *Advancing the embedding transfer paradigm: benchmarking text-integrated protein language models and embedding fusion*. **RECOMB**, Seoul, South Korea, April 2025.

Poster: *Benchmarking text-integrated protein language model embeddings and embedding fusion on diverse downstream tasks*. **Machine Learning in Structural Biology (MLSB) Workshop at NeurIPS**, Vancouver, Canada, December 2024.

AWARDS

Distinguished Graduate Student Fellowship, University of California, San Diego (2024–2025)
- Awarded for research excellence; includes full fellowship support for one quarter and a $3,000 research stipend.

The John Stauffer Scholarship for Academic Merit, Pomona College (2022)
- $12,000 scholarship awarded to one senior annually in the natural sciences.

The Frank Parkhurst Brackett, Jr., and Davida Wark Brackett Prize, Pomona College (2022)
- Awarded to a student as an incentive to excellence in advancing the study of chemistry.

Phi Beta Kappa Award, Pomona College (2022)
- $2,750 award endowed by the Pomona Chapter of Phi Beta Kappa.

REVIEWING

Genome Biology (2026)

BMC Genomics (2026)

Machine Learning for Structural Biology (MLSB) (2025)

NeurIPS Workshop on Machine Learning for Structural Biology (2024)

OUP Bioinformatics (2024, 2026)